

Ata B. Karagoz, Ph.D.

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APPOINTMENTS

The University of Chicago, Chicago, IL

Postdoctoral Scholar, Institute for Mind and Biology (Bakkour Lab)

Aug. 2025 – Present

EDUCATION

Washington University in St. Louis, St. Louis, MO

Ph.D. in Psychological and Brain Sciences

Dissertation: Segmentation and Memory in Interactive, Inferential Events / Advisor: Zachariah M. Reagh, Ph.D.

Aug. 2020 – Aug. 2025

Washington University in St. Louis, St. Louis, MO

M.A. in Psychological and Brain Sciences

Aug. 2020 – Dec. 2022

The University of Texas at Austin, Austin, TX

Bachelor of Science in Neuroscience

Aug. 2014 – May 2018

RESEARCH INTERESTS

How people learn about the world and use what they have learned to guide value-based decisions, studied through the interaction of memory and decision-making and formalized with computational models. Related interests include event memory, reinforcement learning, naturalistic stimuli, and functional neuroimaging.

FELLOWSHIPS, GRANTS, AND AWARDS

Ruth L. Kirschstein NRSA Postdoctoral Fellowship (F32)

2026

Sponsor: Akram Bakkour, Ph.D. / Scored: Impact 30, 20th percentile (pending funding decision)

NINDS Interdisciplinary Training Grant (CCSN)

July 2022 – July 2024

Cognitive, Computational, and Systems Neuroscience; funded by NINDS and the McDonnell Center for Systems Neuroscience

PEER-REVIEWED PUBLICATIONS

Tang, R., **Karagoz, A.B.**, Delarazan, A.I., & Reagh, Z.M. (2026). Active linking through narratives facilitates associative inference. *Proceedings of the National Academy of Sciences*.

Karagoz, A.B., Kool, W.[†], & Reagh, Z.M.[†](2025). Free recall is shaped by inference and scaffolded by event structure. *Communications Psychology*.

Karagoz, A.B., Moran, E.K., Barch, D., Kool, W.[†], & Reagh, Z.M.[†](2025). Evidence for shallow cognitive maps in schizophrenia. *Cognitive, Affective, & Behavioral Neuroscience*.

Karagoz, A.B., Reagh, Z.M.[†], & Kool, W.[†](2024). Construction and use of cognitive maps in model-based control. *Journal of Experimental Psychology: General*.

Karagoz, A.B., Morse, S.J., & Reagh, Z.M. (2023). Cortico-hippocampal networks carry information about characters and their relationships in an extended narrative. *Neuropsychologia*.

Sherrill, K.R., Molitor, R.J., **Karagoz, A.B.**, Atiyam, M., Mack, M.L., & Preston, A.R. (2023). Generalization of cognitive maps across space and time. *Cerebral Cortex*.

† denotes equal contribution.

MANUSCRIPTS UNDER REVIEW & PREPRINTS

Bouffard, N.R., Delarazan, A.I., **Karagoz, A.B.**, Zacks, J.M., & Reagh, Z.M. (2026). Age differences in hippocampal neural timescales during movie-viewing. Preprint: <https://doi.org/10.64898/2026.05.05.723065>.

Morse, S.J., **Karagoz, A.B.**, & Reagh, Z.M. (under review). Event boundaries directionally influence item-level recognition memory. Preprint: <https://psyarxiv.com/h8bj2/>.

MANUSCRIPTS IN PREPARATION

Karagoz, A.B., & Bakkour, A. Learning tunes the weights of a value map but not its resolution.

Karagoz, A.B., Sadek, J., & Bakkour, A. Valence framing gates a memory-based familiarity heuristic in value-guided choice.

Karagoz, A.B., Morse, S.J., Reagh, Z.M.[†], & Kool, W.[†] Complementary roles for PHC and lateral OFC in cognitive map generation.

TALKS & SYMPOSIA

Karagoz, A.B.^{*}, Kool, W., Reagh, Z.M. (2024). Event boundaries and interactive inference shape the content and organization of memory. Nanosymposium talk, Society for Neuroscience (SfN), Chicago, IL.

Karagoz, A.B.^{*}, Kool, W., Reagh, Z.M. (2024). Representations of cognitive maps underlying model-based planning. Symposium talk, Cognitive Neuroscience Society (CNS).

Karagoz, A.B.^{*}, Kool, W., Reagh, Z.M. (2023). Cognitive and neural representations of cognitive maps underlying model-based planning. Nanosymposium talk, Society for Neuroscience (SfN).

POSTER & CO-AUTHORED PRESENTATIONS

Karagoz, A.B.^{*}, Bakkour, A. (2026). Tracking attribute sampling during categorical and relational value inference. Poster, Cognitive Neuroscience Society (CNS).

Karagoz, A.B.^{*}, Kool, W., Reagh, Z.M. (2023). The influence of rule-based event boundaries on item memory. Poster, International Conference of Learning and Memory (LearnMem).

Morse, S.J.^{*}, **Karagoz, A.B.**, Reagh, Z.M. (2023). Event boundaries at encoding influence mnemonic discrimination. Poster, International Conference of Learning and Memory (LearnMem).

Roome, H.E.^{*}, Sherrill, K.R., Nguyen, K.V., **Karagoz, A.B.**, Coughlin, C.A., Preston, A.R. (2022). Medial temporal lobe error signals mediate developmental differences in spatial memory precision. Nanosymposium talk, Society for Neuroscience (SfN).

Sherrill, K.R.^{*}, Roome, H.E., **Karagoz, A.B.**, Long, J.M., Preston, A.R. (2022). Emergence of hippocampal and ventromedial prefrontal cortex context-dependent coding during virtual navigation. Nanosymposium talk, Society for Neuroscience (SfN).

- Karagoz, A.B.***, Reagh, Z.M., Kool, W. (2022). Constructing and using cognitive maps for model-based control. Poster, Reinforcement Learning and Decision Making (RLDM).
- Karagoz, A.B.***, Reagh, Z.M. (2022). Representations of perceptual versus semantic relationships among characters in naturalistic events. Poster, Context and Episodic Memory Symposium (CEMS).
- Reagh, Z.M.* , Morse, S.J., Fishman, R., Angulo-Lopera, S., **Karagoz, A.B.**, & Delarazan, A.I. (2022). Event boundaries at encoding influence mnemonic discrimination. Poster, Cognitive Neuroscience Society (CNS).
- Delarazan, A.I.* , **Karagoz, A.B.***, Montchal, M.E., Yassa, M.A., Ranganath, C., Reagh, Z.M. (2021). Hippocampal and entorhinal contributions to naturalistic event context reinstatement. Virtual poster, Society for Neuroscience (SfN).
- Karagoz, A.B.***, Reagh, Z.M., Kool, W. (2021). The construction and use of cognitive maps in model-based control. Virtual poster, Psychonomic Society Annual Meeting.
- Karagoz, A.B.***, Reagh, Z.M. (2021). Decoding perceptual and semantic relatedness among characters in naturalistic events. Virtual poster, Cognitive Neuroscience Society (CNS).

* denotes presenting author.

RESEARCH EXPERIENCE

Bakkour Lab, The University of Chicago Aug. 2025 – Present
Postdoctoral Scholar / Supervisor: Akram Bakkour, Ph.D.

- Design behavioral and fMRI studies of memory-guided, value-based decision-making.
- Collect and analyze fMRI datasets; develop computational models of choice behavior.

Complex Memory Lab, Washington University in St. Louis Aug. 2020 – Aug. 2025
Graduate Research Assistant / Supervisor: Zachariah Reagh, Ph.D.

- Designed and ran behavioral and fMRI studies of event memory and model-based control.
- Built computational models linking event structure, inference, and recall.

Preston Lab, The University of Texas at Austin Jan. 2018 – Aug. 2020
Lab Manager / Supervisor: Alison Preston, Ph.D.

- Coordinated data collection and analysis across memory-integration and navigation projects.
- Scheduled sessions and built the participant pool for behavioral and scanning studies.

Preston Lab, The University of Texas at Austin Nov. 2015 – Jan. 2018
Undergraduate Research Assistant / Supervisor: Alison Preston, Ph.D.

- Assisted in fMRI scanning studies; wrote MATLAB analysis scripts for ongoing projects.
- Designed a study and developed its stimulus-presentation code in MATLAB.

TEACHING & MENTORING

Graduate Teaching Assistant, Washington University in St. Louis Jan. 2023 – May 2023
Course: Computational Cognitive Science (upper-level undergraduate)

- Wrote and graded Python programming assignments; led weekly review sessions.

Teaching Assistant, Washington University in St. Louis Jan. 2022 – May 2022
Course: Neurobiology of Learning and Memory

- Delivered a guest lecture on spatial navigation; graded assignments and led exam review sessions.

Teaching Assistant, *Washington University in St. Louis*
Course: Human Learning and Memory

Sep. 2021 – Dec. 2021

- Wrote and graded quizzes; led exam review sessions.

Mentoring — Master’s Students

Oct. 2025 – Present

- Jalal Sadek (**The University of Chicago**) — co-author on manuscript in preparation.

Mentoring — Undergraduate Research Assistants
Complex Memory Lab, Washington University in St. Louis

June 2018 – Present

- Ron Fishman, Jacob Tartakovsky, Sofia Angulo-Lopera
Preston Lab, The University of Texas at Austin
- Jaida McKee, Doru Gucer, Katherine Vasquez

SERVICE & OUTREACH

Ad-Hoc Peer Reviewer

2022 – Present

- *Neuron* (co-review), *Nature Communications*, *eLife* (co-review), *Cerebral Cortex*, *Cognition*, *Communications Psychology*

Psychology Outreach with Elementary Schoolers

Sep. 2020 – Dec. 2020

- Taught introductory psychology over Zoom; designed remote perception demonstrations.

GitHub Clinic, *Complex Memory Lab*

Sep. 2020

- Ran a lab-wide workshop on version control and reproducible git practice.

Open Science

- Analysis code and task materials publicly archived: [TODO: OSF / GitHub links].

PROFESSIONAL MEMBERSHIPS

- Cognitive Neuroscience Society, Postdoctoral Member
- Psychonomic Society, Member

SKILLS

- **Programming:** Python, MATLAB, bash scripting
- **Computational Modeling:** reinforcement learning and model-based control models, hierarchical Bayesian parameter estimation, PyTorch, JAX
- **fMRI Preprocessing & Analysis:** fMRIPrep, FSL/FEAT, ANTs, multivariate pattern analysis
- **Reproducible Workflows:** git/GitHub, Jupyter, Docker/Singularity
- **High-Performance Computing:** Texas Advanced Computing Center, WashU Center for High Performance Computing, UChicago Research Computing Center
- **Languages:** English (native), Turkish (native), German (basic)